01 Mouth, pharynx and salivary glands

Team: _____

Basic function (this is all you need): Food enters, is chewed, mixed with saliva, and swallowed.

STRUCTURES TO DRAW AND LABEL

- Lips, cheeks, vestibule, oral cavity proper
- Hard palate (palatine processes of the maxillae, horizontal plates of the palatine bones) and soft palate with the uvula
- Palatoglossal and palatopharyngeal arches with the palatine tonsils between them
- Tongue: body, root, lingual frenulum; papillae: filiform, fungiform, foliate, circumvallate (vallate)
- Teeth: 20 deciduous, 32 permanent; incisors, canines, premolars, molars; crown, neck, root
- Tooth layers: enamel, dentin, pulp cavity, cementum, periodontal ligament
- Parotid gland and parotid duct (opens opposite the upper second molar)
- Submandibular gland and duct (opens at the sublingual caruncle beside the frenulum)
- Sublingual gland (many small ducts along the sublingual fold)
- Pharynx: nasopharynx, oropharynx, laryngopharynx; pharyngeal constrictor muscles

NERVES AND BLOOD SUPPLY

- Parasympathetic to the parotid: CN IX (glossopharyngeal) through the otic ganglion
- Parasympathetic to the submandibular and sublingual glands: CN VII (facial) through the submandibular ganglion
- Sympathetic to all salivary glands: superior cervical ganglion
- Tongue sensation: anterior two thirds general sensation CN V3 (lingual nerve), taste CN VII (chorda tympani); posterior third both by CN IX
- Tongue muscles: CN XII (hypoglossal), except palatoglossus (CN X)
- Pharynx: pharyngeal plexus (CN IX and CN X)
- Enteric plexuses: submucosal (in the submucosa) and myenteric (between the circular and longitudinal muscle). Present along the whole tube.
- Artery: External carotid branches: facial, lingual and maxillary arteries

THE WALL UNDER THE MICROSCOPE

- Oral lining: nonkeratinized stratified squamous epithelium. Keratinized on the gingiva and hard palate
- Tongue: skeletal muscle core running in three planes, covered by stratified squamous epithelium with papillae
- Taste buds on fungiform, foliate and circumvallate papillae (not filiform)
- Salivary glands: serous acini (dark, round nuclei, zymogen granules), mucous acini (pale, flattened basal nuclei), serous demilunes on mixed glands
- Parotid: almost all serous. Submandibular: mixed, mostly serous. Sublingual: mixed, mostly mucous
- Oropharynx and laryngopharynx: stratified squamous. Nasopharynx: pseudostratified ciliated columnar

CELLS AND WHAT THEY MAKE

Serous cells (salivary glands)	Watery fluid carrying salivary amylase and lysozyme
Mucous cells (salivary glands)	Mucin, which becomes mucus
Duct cells (striated ducts)	Adjust the salt content of saliva
Gustatory (taste) cells	Detect taste; found in taste buds
Odontoblasts	Dentin (for the life of the tooth)
Ameloblasts	Enamel, only before the tooth erupts

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Analogy: three faucets

Parotid in the cheek runs pure water (all serous). Submandibular under the jaw is mostly water with some slime (mixed, mostly serous). Sublingual under the tongue is mostly slime (mixed, mostly mucous). The lower the gland, the slimier the saliva.

Mnemonic: Every Dentist Pulls Carefully

Enamel, Dentin, Pulp, Cementum. The first three run outside to inside through the crown; cementum coats the root. Tongue taste: seven for the tip, nine for the back (CN VII anterior two thirds, CN IX posterior third).